Equipment Frontend Implementation Guide

Base API path: `/api/inventory`

Auth and Roles

- Public/optional: equipment list, equipment by id, categories list
- Team member or higher: create, update, checkout, return
- Mentor or higher: delete

Data Models

Equipment

- **name** string [2–255], required
- **description** string [≤2000]
- **categoryId** ObjectId (ref `EquipmentCategory`), required
- **modelNumber** string [≤100]
- **serialNumber** string [≤100], unique, sparse
- **manufacturer** string [≤100]
- **purchaseDate** Date
- **purchasePrice** number (≥0)
- **currentQuantity** int (≥0), default 0
- **minQuantity** int (≥0), default 0
- **maxQuantity** int (≥0)
- **location** string [≤255]
- **status** enum: `available | low_stock | out_of_stock | maintenance | retired`
- **imageUrl** string [≤500]
- **specifications** object
- **maintenanceSchedule** string [≤1000]
- **lastMaintenanceDate** Date
- **nextMaintenanceDate** Date
- **checkouts** array of checkout subdocuments

- **createdAt**, **updatedAt** timestamps

Checkout Subdocument

- **userId** ObjectId (User), required
- **quantity** int (≥1), required
- **checkoutDate** Date (default now)
- **expectedReturnDate** Date, required
- **actualReturnDate** Date
- **checkoutNotes** string [≤1000]
- **returnNotes** string [≤1000]
- **status** enum: `checked_out | returned | overdue | lost`

Category

- **name** string [2–100], required
- **description** string (optional)
- **parentCategoryId** ObjectId (optional)

Endpoints

GET `/equipment`

Auth: optional

Query: `categoryId`, `status`, `q`, `limit` (1–100), `skip` (≥0)

Response: `{ success, data: { items, pagination: { total, limit, skip, hasMore } } }`

GET `/equipment/:id`

Auth: optional

Response: `{ success, data: { item } }`

POST `/equipment`

Auth: team_member+

Body: name (2–255), categoryId, currentQuantity?, status?, other fields accepted

Response: `201 { success, message, data: { item } }`

PUT `/equipment/:id`

Auth: team_member+

Body: any updatable fields, including `imageUrl`

Response: `{ success, message, data: { item } }`

DELETE `/equipment/:id`

Auth: mentor+

Response: `{ success, message }`

POST `/equipment/:id/checkout`

Auth: team_member+

Body: quantity (default 1), expectedReturnDate (default now+7d)

Response: `201 { success, message, data: { currentQuantity } }`

PUT `/equipment/:id/return`

Auth: team_member+

Body: checkoutId? (if omitted, returns last active for user)

Response: `{ success, message, data: { currentQuantity } }`

GET `/categories`

Auth: optional

Response: `{ success, data: { items: Category[] } }`

POST `/categories`

Auth: team_member+

Body: name (2–100), description?, parentCategoryId?

Response: `201 { success, message, data: { item } }`

PUT `/categories/:id`

Auth: team_member+

Response: `{ success, message, data: { item } }`

DELETE `/categories/:id`

Auth: mentor+

Response: `{ success, message }`

UI Guidance

List Page

- Filters: Category, Status, Search, pagination
- Columns: Name, Category, Status, Current qty, Location, Updated at, Actions
- Role-based actions: View, Edit (team_member+), Delete (mentor+)

Detail Page

- Overview, Inventory, Maintenance, Specifications, Checkout history
- Actions: Checkout, Return, Edit/Delete by role

Create/Edit Form

- Required: name, categoryId
- Client validation mirrors server rules

Example Requests

```
GET /api/inventory/equipment?q=printer&status=available&limit=20&skip=0
```

```
GET /api/inventory/equipment/:id
```

```
POST /api/inventory/equipment
```

```
PUT /api/inventory/equipment/:id
```

```
// Example: Update imageUrl
PUT /api/inventory/equipment/:id
Authorization: Bearer <token>
Content-Type: application/json

{
  "imageUrl": "https://cdn.example.com/inventory/equipment-123.jpg"
}
```

```
DELETE /api/inventory/equipment/:id
```

```
POST /api/inventory/equipment/:id/checkout
```

```
PUT /api/inventory/equipment/:id/return
```

```
GET /api/inventory/categories
```

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